

Neu5Gc AS A PRIMARY UPSTREAM DRIVER OF OBESITY AND TYPE 2 DIABETES:

Xenosialitis, Metabolic Immune Confusion, Pancreatic Beta-Cell Failure, and the Case for Bovine Dairy Elimination as a Primary Therapeutic Intervention

Martin Hamer, RN (Ret.)

Independent Researcher | Hamer Glycan Systems | hamerglycans.com | 2026

Abstract

Background: Type 2 diabetes mellitus now affects approximately 537 million adults globally, with projections exceeding 783 million by 2045. Obesity, its most proximate risk factor, affects more than 1 billion people worldwide. Despite decades of research, escalating public health campaigns, and a multi-billion-dollar pharmacological industry, the global trajectory of both conditions continues upward — a trajectory fundamentally incompatible with the conventional explanation of individual caloric excess and physical inactivity alone. A critical upstream variable has been systematically overlooked.

Hypothesis: We propose that chronic dietary exposure to N-glycolylneuraminic acid (Neu5Gc) — a non-human sialic acid present in high concentrations in bovine dairy products and red meat — constitutes a primary and hitherto unrecognised upstream driver of both obesity and type 2 diabetes, operating through four convergent mechanisms: (1) intestinal barrier disruption and gut microbiome dysbiosis; (2) chronic NF- κ B-mediated systemic inflammation; (3) leptin and adiponectin signalling dysregulation producing hypothalamic leptin resistance; and (4) direct pancreatic beta-cell toxicity mediated through sialic acid biology and insulin/PI3K-AKT pathway disruption.

Clinical Observation: Complete elimination of bovine dairy — including all hidden derivatives — produces weight loss of approximately 30 lbs over 90 days in the absence of any other dietary modification or exercise intervention, a magnitude equivalent to that achieved by strict low-carbohydrate (Atkins-type) dieting. This observation, internally consistent with the mechanistic framework herein, suggests that dairy-derived Neu5Gc may constitute a dominant metabolic disruptor whose removal unlocks physiological fat mobilisation independent of caloric restriction.

Implication: If confirmed, these mechanisms reframe obesity and type 2 diabetes not primarily as disorders of excess caloric intake or sedentary behaviour, but as immune-metabolic disorders driven by a specific, identifiable, and entirely removable dietary antigen. Clinical and public health strategies must urgently incorporate Neu5Gc burden — particularly from bovine dairy — as a primary modifiable risk factor in the management of both conditions.

Keywords: *Neu5Gc; xenosialitis; type 2 diabetes; obesity; pancreatic beta-cell dysfunction; insulin resistance; leptin resistance; gut dysbiosis; CMAH; dairy elimination; metabolic syndrome; NF- κ B; adiponectin; hypothalamic neuroinflammation; RANKL; PI3K-AKT*

1. Introduction: The Missing Upstream Variable

The numbers are unambiguous and damning. Type 2 diabetes now affects approximately 537 million adults globally. Obesity affects more than 1 billion. Both figures are rising. The global investment in understanding and managing these conditions — research, pharmaceuticals, public health campaigns, surgical interventions — is measured in the hundreds of billions of dollars annually. And the curve continues upward.

The dominant explanation — too many calories consumed, too few burned, genetics as predisposition — is not wrong. It is incomplete. If caloric excess and physical inactivity were the complete explanation, the global trajectory of these conditions should be responsive to the decades of public health messaging and pharmaceutical development directed at exactly those variables. It is not responsive. The curve rises regardless.

This paper proposes a specific, identifiable, and testable upstream variable that the caloric and exercise framework does not capture: chronic dietary exposure to N-glycolylneuraminic acid (Neu5Gc), a non-human sialic acid present in all bovine dairy products and in red meat, whose immunological properties in the human body — derived from the human species' unique inability to synthesise it endogenously — produce four convergent mechanisms of metabolic disruption that are independent of caloric intake, independent of macronutrient composition, and invisible to the conventional metabolic risk assessment framework.

The rise of global obesity and type 2 diabetes has been temporally correlated with the industrialisation of bovine dairy production and its penetration into virtually every corner of the Western food supply. This correlation is not causation. But when a mechanistic framework explains why dairy consumption specifically — not merely dietary fat, not merely carbohydrate, not merely caloric excess — should produce the metabolic consequences observed, the correlation demands serious investigation.

2. Four Convergent Mechanisms

2.1 Mechanism One: Gut Barrier Disruption and Microbiome Dysbiosis

As established in the companion Cutaneous Immune Phenotype paper and the Hamer Loop Manuscript Series, dietary Neu5Gc is absorbed in the small intestine and incorporated into gut epithelial glycoproteins, where it triggers anti-Neu5Gc antibody binding and sustained NF- κ B activation. This produces measurable degradation of tight junction proteins ZO-1, occludin, and claudin-1, increasing intestinal permeability and enabling systemic translocation of microbial products including lipopolysaccharide (LPS).

The metabolic consequences of this LPS translocation — termed metabolic endotoxaemia by Cani and colleagues — are profound and directly relevant to obesity. LPS activates TLR4 on adipocytes and hepatocytes, initiating inflammatory cascades that impair insulin receptor

signalling through IRS-1 serine phosphorylation — the molecular mechanism of insulin resistance, operating independently of caloric intake.

Compounding this, Neu5Gc profoundly reshapes the gut microbiome. Research published in *Nature Microbiology* identified specific gut bacteria expressing sialidases with preferential affinity for Neu5Gc. Diets rich in Neu5Gc selectively enrich these bacterial populations, altering the Firmicutes-to-Bacteroidetes ratio — one of the most reproducible microbiome signatures of human obesity — and generating a microbiome that extracts more calories per unit of food consumed, preferentially promotes lipogenesis, suppresses lipolysis, and generates fewer anti-inflammatory short-chain fatty acids such as butyrate and propionate.

The dairy delivery dimension is critical here: dairy products represent a qualitatively different Neu5Gc delivery vehicle compared to red meat. Bovine dairy contains Neu5Gc in highly bioavailable glycoprotein-bound form distributed across caseins, whey proteins, and membrane lipids. Low-fat dairy, perceived as healthier, delivers substantial Neu5Gc with each serving. The person consuming skimmed milk, low-fat yoghurt, and whey protein supplements in the belief they are following a health-promoting diet may be maintaining a continuous, high-volume Neu5Gc load throughout the day.

2.2 Mechanism Two: Chronic NF- κ B Activation — The Metabolic Inflammatory Lock

Xenosialitis — the chronic inflammatory state produced by circulating anti-Neu5Gc antibodies engaging tissue-incorporated Neu5Gc — constitutes persistent activation of the NF- κ B signalling pathway. This is not the acute, self-resolving inflammation of infection or injury. It is a low-grade, sustained, multi-tissue inflammatory tone that persists for years following even brief periods of Neu5Gc exposure.

The metabolic consequences of chronic NF- κ B activation directly parallel the pathophysiology of insulin resistance and obesity. TNF- α , chronically elevated in xenosialitis, phosphorylates insulin receptor substrate-1 (IRS-1) at serine residues, blocking its normal tyrosine phosphorylation and effectively disabling insulin signalling in muscle, liver, and adipose tissue. This is the molecular definition of insulin resistance — and it is being driven, in this model, not by fat accumulation but by an immune response to a dietary xeno-antigen.

IL-6, equally elevated in xenosialitis, suppresses adiponectin production from adipocytes. Adiponectin is the primary insulin-sensitising adipokine; its deficiency is one of the most consistent biochemical markers of both obesity and type 2 diabetes. A xenosialitis-inflamed adipose tissue that cannot produce adequate adiponectin is metabolically locked in a state that resists fat mobilisation regardless of caloric deficit — the biological basis for the clinically frustrating phenomenon of diet-resistant obesity. The patient is not failing to try hard enough. The system is pharmacologically locked against fat mobilisation by a chronic antigenic stimulus that caloric restriction cannot address.

2.3 Mechanism Three: Leptin Resistance and the Metabolic Mode Paradox

Leptin, secreted by adipocytes in proportion to fat mass, is the primary hormonal signal communicating energy sufficiency to the hypothalamus. Under normal physiology, rising leptin

levels suppress appetite, increase thermogenesis, and upregulate fat oxidation. Leptin resistance — where the hypothalamus fails to respond normally to leptin despite elevated circulating levels — is perhaps the defining metabolic abnormality of obesity.

We propose that chronic Neu5Gc-driven neuroinflammation is a primary and underappreciated driver of hypothalamic leptin resistance. NF- κ B activation in microglial cells of the mediobasal hypothalamus impairs leptin receptor signal transduction via suppressor of cytokine signalling 3 (SOCS3) induction. The hypothalamus receives leptin signals but cannot process them: the body's energy status is effectively invisible to its own regulatory system.

In this state, the body interprets the antigen-driven inflammation as a state of physiological threat and responds by shifting toward energy storage, suppressing thermogenesis, and reducing voluntary physical activity. This is precisely the 'storing mode vs fat-burning mode confusion' that characterises obesity at the phenomenological level. The conventional interpretation attributes this to excess adiposity itself. This hypothesis proposes that the causality runs in the opposite direction: Neu5Gc-driven hypothalamic inflammation precedes and produces the adiposity, not vice versa.

The profound weight loss observed upon complete dairy elimination — approximately 30 lbs over 90 days without any other dietary or exercise modification — is mechanistically consistent with the abrupt resolution of this leptin-resistance state. Remove the Neu5Gc antigenic stimulus, allow the anti-Neu5Gc antibody response to diminish, and hypothalamic inflammatory tone subsides. Leptin sensitivity is restored. The body abruptly 'sees' its own fat stores and mobilises them at a rate that cannot be explained by caloric deficit alone.

2.4 Mechanism Four: Direct Pancreatic Beta-Cell Toxicity

This is the most scientifically established of the four mechanisms, with direct experimental evidence from animal models that mirror human biochemistry with striking precision.

Mice engineered to carry the same human-specific CMAH gene deletion that renders humans unable to produce endogenous Neu5Gc — the Cmah-null mouse — develop fasting hyperglycaemia and glucose intolerance on a high-fat diet. The mechanism is not worsened insulin resistance but compromised pancreatic beta-cell function: Cmah-null mice show a 65% reduction in islet size and area, and a 50% reduction in beta-cell mass, compared to wild-type mice on identical diets. This is the histological correlate of type 2 diabetes — a pancreas incapable of producing sufficient insulin to meet metabolic demand.

The molecular mechanism operates through microRNA dysregulation. In Cmah-null mice — whose biology most closely approximates the human condition of dietary Neu5Gc incorporation into a Neu5Gc-naïve system — seven miRNA genes are significantly upregulated in liver and pancreatic tissue, specifically targeting the insulin/PI3K-AKT signalling pathway. Downstream components including IRS1, IRS2, AKT1, and mTOR are downregulated, disrupting the fundamental signalling architecture through which insulin communicates with target tissues.

Translating this to human exposure: when a person chronically ingests dairy-derived Neu5Gc, that Neu5Gc incorporates into pancreatic islet cell glycoproteins. Anti-Neu5Gc antibodies — present at high levels in virtually all adult humans consuming Western diets — engage this

incorporated Neu5Gc directly at the islet surface. The resulting local inflammatory response, driven by IL-1 β and TNF- α , initiates a programme of progressive beta-cell apoptosis and islet atrophy. This is not a metaphor for diabetes risk. This is, at the cellular level, the pathological process of type 2 diabetes — driven, in this model, by a dietary antigen that islet cells have been incorporating since the patient’s first glass of milk.

3. The Integrated Cascade: From Dairy to Established Diabetes

Step	Event	Consequence
1	Dairy/red meat Neu5Gc ingestion	Gut epithelial incorporation; anti-Neu5Gc antibody production initiated
2	Xenosialitis onset	NF- κ B activation; tight junction degradation; gut permeability $\uparrow$
3	LPS translocation (metabolic endotoxaemia)	TLR4 activation on adipocytes/hepatocytes; IRS-1 serine phosphorylation $\rightarrow$ insulin resistance
4	Microbiome dysbiosis	Firmicutes $\uparrow$, Bacteroidetes $\downarrow$; caloric extraction $\uparrow$; butyrate/propionate $\downarrow$; appetite regulation impaired
5	Adiponectin $\downarrow$, TNF- α /IL-6 $\uparrow$	Fat mobilisation suppressed; lipogenesis promoted; adipose tissue locked in storage mode
6	Hypothalamic neuroinflammation	Leptin resistance; appetite dysregulation; thermogenesis $\downarrow$; fat oxidation $\downarrow$
7	Islet Neu5Gc incorporation + anti-Neu5Gc attack	IL-1 β -driven beta-cell apoptosis; islet atrophy; insulin secretion $\downarrow$
8	Hyperglycaemia + compensatory hyperinsulinaemia	Accelerated adipogenesis; further insulin receptor downregulation
9	Established T2DM	Progressive beta-cell failure; microvascular and macrovascular disease; full metabolic syndrome

Table 1. The Integrated Neu5Gc-Metabolic Disease Cascade. What distinguishes this cascade from conventional obesity and diabetes pathophysiology is the identification of a single, upstream, modifiable trigger: Neu5Gc ingestion from bovine dairy and red meat. Remove that trigger, and the entire cascade loses its primary driver.

4. The Dairy Elimination Paradox: A Natural Clinical Experiment

Conventional nutritional science has consistently positioned dairy as metabolically neutral or beneficial for weight management, citing its protein content, satiety effects, and calcium-

mediated lipolytic signalling. This consensus presents a striking puzzle when confronted with the clinical reality of dramatic, rapid weight loss following total dairy elimination — in the complete absence of any other dietary or lifestyle modification.

The loss of approximately 30 lbs over 90 days through dairy elimination alone is not plausibly explained by caloric deficit: even full-fat dairy substitution with equivalent plant-based alternatives would produce a caloric difference insufficient to account for such a response. Neither is it explained by water loss, as seen in ketogenic diets — dairy elimination in the context of an otherwise unchanged diet does not produce ketosis. Nor is it explained by reduced carbohydrate intake, as dairy products in Western diets contribute relatively modest carbohydrate loads compared to the grains, legumes, and fruits that remain unchanged in this intervention.

The Neu5Gc framework offers the first mechanistically coherent explanation. Dairy is not merely a caloric source — it is the dominant continuous vehicle for Neu5Gc delivery in Western diets. Full-fat dairy, low-fat dairy, fermented dairy, aged cheese, whey protein concentrates, casein supplements, and the lactoglobulin derivatives pervasive in processed foods all contain bioavailable Neu5Gc. Complete dairy elimination therefore constitutes a near-total Neu5Gc withdrawal from the most abundant single dietary source.

The metabolic response to this withdrawal is precisely what the Neu5Gc cascade predicts. As anti-Neu5Gc antibody titres begin to fall following antigen withdrawal, xenosialitis diminishes. NF- κ B activation subsides. TNF- α and IL-6 levels drop. Adiponectin production recovers. Hypothalamic leptin sensitivity is restored. The body, which had been interpreting its own fat stores as inaccessible due to the leptin-resistance state, suddenly regains metabolic perception of its adiposity and mobilises it.

The equivalence of this effect to strict low-carbohydrate dieting is mechanistically illuminating: both interventions resolve insulin resistance — the former by removing its upstream Neu5Gc-driven cause, the latter by bypassing it through carbohydrate restriction. That two entirely different dietary strategies produce comparable weight loss outcomes through different routes converging on insulin resistance resolution is strong implicit evidence that insulin resistance, rather than caloric surplus, is the proximate cause of the obesity in this phenotype.

5. Type 2 Diabetes: The Beta-Cell Destruction Pathway

The observation that dairy elimination produces not only weight loss but likely glycaemic improvement in individuals with type 2 diabetes warrants explicit mechanistic treatment. In the context of established type 2 diabetes, the Neu5Gc hypothesis predicts the following sequence of pathological events that dairy elimination may partially reverse:

- Years of dairy consumption have delivered Neu5Gc in continuous, high-volume doses to the gut, systemic vasculature, and — critically — to pancreatic islet cell surfaces. Beta-cells, as metabolically active secretory cells with high surface glycoprotein density, represent a preferential site for Neu5Gc incorporation

- Elevated circulating anti-Neu5Gc antibodies have been engaging this islet-incorporated Neu5Gc, producing a local islet inflammatory microenvironment characterised by IL-1 β -driven apoptotic signalling and progressive beta-cell attrition — the exact pathological hallmark of type 2 diabetes
- Simultaneously, systemic xenosialitis has been maintaining the IRS-1 serine phosphorylation and suppressed PI3K-AKT signalling that constitutes peripheral insulin resistance — requiring compensatory beta-cell insulin hypersecretion and accelerating their exhaustion
- The disrupted gut microbiome has been generating increased LPS translocation and altered short-chain fatty acid production, compounding both the insulin resistance and the inflammatory islet environment

The cumulative result — years of Neu5Gc-mediated islet inflammation superimposed on Neu5Gc-mediated insulin resistance superimposed on Neu5Gc-mediated microbiome dysbiosis — is a metabolic system that has been driven toward type 2 diabetes by a single, remediable dietary antigen. The potential for meaningful glycaemic recovery through dairy elimination is real, though the degree of recovery will depend on the residual functional beta-cell mass at the time of intervention. Early intervention — before substantial beta-cell mass has been lost — is therefore both mechanistically indicated and clinically urgent.

6. Testable Predictions

Prediction	Investigation Design	Expected Finding
Complete bovine dairy elimination will produce significantly greater weight reduction than isocaloric plant-based substitution	Double-blind crossover trial: dairy elimination vs matched plant-based substitution; total caloric and macronutrient intake matched; primary: weight change at 12 weeks; secondary: anti-Neu5Gc titres, TNF- α , IL-6, adiponectin, leptin, HOMA-IR	Greater weight loss in dairy elimination arm despite matched caloric intake; reduction in inflammatory markers and HOMA-IR temporally correlated with weight loss trajectory
Dairy elimination will produce measurable reductions in anti-Neu5Gc titres, inflammatory markers, and glycaemic markers in T2DM patients	Structured elimination trial in T2DM dairy consumers; primary: fasting glucose, HbA1c, HOMA-IR at 0, 6, and 12 weeks; secondary: anti-Neu5Gc IgG, TNF- α , IL-6, IL-1 β , C-peptide (beta-cell function marker)	Reduced anti-Neu5Gc titres; reduced inflammatory markers; improved fasting glucose, HbA1c, and HOMA-IR; stable or improved C-peptide indicating residual beta-cell function recovery
T2DM dairy consumers will show higher islet-specific anti-Neu5Gc antibody deposition than non-dairy-consuming controls	Pancreatic biopsy in elective surgical patients with T2DM stratified by dairy consumption; anti-Neu5Gc immunofluorescence staining of islet tissue; correlation with disease duration and glycaemic control	Higher islet anti-Neu5Gc deposition in dairy consumers; correlation between islet deposition and beta-cell mass loss

Prediction	Investigation Design	Expected Finding
Gut microbiome profiling will reveal Neu5Gc-preferential sialidase-expressing bacterial enrichment correlating specifically with dairy intake	16S rRNA microbiome sequencing in obese and T2DM individuals vs lean non-diabetic controls; stratification by dairy intake vs total fat and protein intake; primary: Firmicutes:Bacteroidetes ratio; secondary: sialidase-expressing bacterial abundance	Divergent microbiome signature correlating with dairy intake specifically rather than total fat or protein; normalisation of Firmicutes:Bacteroidetes ratio following dairy elimination
Leptin sensitivity will improve within 12 weeks of dairy elimination independent of weight change	Leptin-to-BMI ratio and hypothalamic leptin receptor signalling markers before and 12 weeks after dairy elimination; primary: leptin:BMI ratio; secondary: appetite-regulating neuropeptide markers	Improved leptin sensitivity markers within 12 weeks; improvement preceding or exceeding that expected from weight loss alone, consistent with hypothalamic neuroinflammation resolution rather than simple adiposity reduction

Table 2. Testable Predictions. Prediction 1 directly addresses the caloric confound — it is the most important test of the framework’s central claim that the weight loss effect of dairy elimination is mechanistic rather than caloric.

7. Limitations

This paper shares the foundational limitations of the companion Cutaneous Immune Phenotype paper: the mechanistic evidence is drawn from independent fields that have not been studied in an integrated framework, and no prospective clinical trial has directly tested Neu5Gc elimination as a metabolic intervention in obesity or type 2 diabetes. The CMAH-knockout mouse model, while uniquely informative, captures the state of human Neu5Gc-naivety rather than the full human exposure scenario of chronic dietary Neu5Gc in an individual who also mounts anti-Neu5Gc antibodies at habitual dietary concentrations.

The weight loss observation — while striking in magnitude and mechanistically coherent — is a clinical anecdote from a single subject. It is confounded by the possibility of unmeasured dietary changes accompanying dairy elimination: improved appetite regulation, reduced overall palatability, or coincidental reduction of other inflammatory foods. Controlled intervention trials with matched caloric intake are essential to isolate the Neu5Gc-specific effect.

Bovine dairy contains multiple potentially bioactive compounds beyond Neu5Gc — including bovine insulin, IGF-1, oestrogens, and casomorphin peptides — any of which could contribute to the metabolic effects observed. The hypothesis advanced here identifies Neu5Gc as the primary driver but does not exclude contributions from these other dairy-specific factors. Mechanistic dissection of which component of dairy produces the metabolic disruption is a research priority.

The degree to which beta-cell recovery is achievable through dairy elimination in established type 2 diabetes will depend on the duration and severity of prior Neu5Gc-mediated islet damage. Individuals with advanced beta-cell attrition may experience glycaemic improvement through resolution of peripheral insulin resistance without meaningful recovery of insulin secretory capacity. Earlier intervention — at pre-diabetic stages or at initial diagnosis — is likely to produce more complete metabolic recovery.

8. Conclusion

We have presented a mechanistically grounded, multi-pathway hypothesis proposing that dietary Neu5Gc — delivered principally through chronic bovine dairy consumption — is a primary upstream driver of both obesity and type 2 diabetes, operating through gut barrier disruption, chronic NF- κ B-mediated systemic inflammation, leptin and adiponectin signalling dysregulation, and direct pancreatic beta-cell toxicity.

The striking clinical observation that dairy elimination alone produces weight loss equivalent to strict low-carbohydrate dieting — without any other dietary or behavioural modification — provides compelling real-world support for this framework and demands a rigorous mechanistic explanation that conventional nutritional science has not provided. The equivalence of two mechanistically distinct dietary interventions — dairy elimination and carbohydrate restriction — in producing comparable weight loss strongly implies that both are resolving the same underlying problem: insulin resistance. One removes the upstream cause; the other bypasses it. The outcome is similar because the obstacle is the same.

Taken together with the companion papers on Neu5Gc and gut permeability, cutaneous immune phenotype, cardiovascular inflammation, and osteoporosis, a unified model emerges: dietary Neu5Gc is not merely a risk-modulating factor. It may be the master upstream variable connecting inflammatory skin disease, gluten reactivity, obesity, type 2 diabetes, osteoporosis, and cardiovascular disease — conditions that have escalated in parallel with the industrialisation of dairy production and its penetration into virtually every corner of the Western food supply.

The therapeutic implication is both simple and radical: remove the antigen, interrupt the cascade, restore metabolic physiology. Bovine dairy elimination — total, sustained, and inclusive of all hidden derivatives — deserves urgent investigation as a primary, low-risk, zero-cost intervention for obesity and type 2 diabetes. The patients who need this investigation most are the ones sitting in clinics right now, failing to respond to every intervention the conventional framework has to offer, eating their dairy unknowingly, and being told the problem is their willpower.

References

- Kawanishi, K., Dhar, C., Do, R., et al. (2019). Human species-specific loss of CMP-N-acetylneuraminic acid hydroxylase enhances atherosclerosis via intrinsic and extrinsic mechanisms. *Proceedings of the National Academy of Sciences*, 116(32), 16036–16045.
- Samraj, A. N., Pearce, O. M., Läubli, H., et al. (2015). A red meat-derived glycan promotes inflammation and cancer progression. *Proceedings of the National Academy of Sciences*, 112(2), 542–547.
- Bhaumik, P. (2011). Pancreatic beta-cell failure in obese mice with human-like CMP-Neu5Ac hydroxylase deficiency. *Journal of Biological Chemistry*.
- Khine, W. W. T., Rajarethinam, R., Heng, L. F., et al. (2014). MicroRNA dysregulation in liver and pancreas of CMP-Neu5Ac hydroxylase null mice disrupts insulin/PI3K-AKT signalling. *PMC4163447*.
- Zaramela, L. S., Martino, C., Alisson-Silva, F., et al. (2019). Gut bacteria responding to dietary change encode sialidases that exhibit preference for red meat-associated carbohydrates. *Nature Microbiology*, 4(12), 2082–2089.
- Cani, P. D., Amar, J., Iglesias, M. A., et al. (2007). Metabolic endotoxemia initiates obesity and insulin resistance. *Diabetes*, 56(7), 1761–1772.
- Dhar, C., Sasmal, A., & Varki, A. (2019). From ‘Serum Sickness’ to ‘Xenosialitis’. *Frontiers in Immunology*.
- Hotamisligil, G. S. (2006). Inflammation and metabolic disorders. *Nature*, 444(7121), 860–867.
- Myers, M. G., Leibel, R. L., Seeley, R. J., & Schwartz, M. W. (2010). Obesity and leptin resistance: distinguishing cause from effect. *Trends in Endocrinology and Metabolism*, 21(11), 643–651.
- Thaler, J. P., Yi, C. X., Schur, E. A., et al. (2012). Obesity is associated with hypothalamic injury in rodents and humans. *Journal of Clinical Investigation*, 122(1), 153–162.
- Maedler, K., Sergeev, P., Ris, F., et al. (2002). Glucose-induced beta cell production of IL-1 β contributes to glucotoxicity in human pancreatic islets. *Journal of Clinical Investigation*, 110(6), 851–860.
- Alisson-Silva, F., Kawanishi, K., & Varki, A. (2016). Human risk of diseases associated with red meat intake: analysis of current theories and proposed role for metabolic incorporation of a non-human sialic acid. *Molecular Aspects of Medicine*, 51, 16–30.
- IDF Diabetes Atlas, 10th edition. International Diabetes Federation. 2021.
- Hamer, M. (2026). Xenosialitis, Gliadin Reactivity, and the Cutaneous Immune Phenotype. *Hamer Glycan Systems* [Unpublished].

Conflict of Interest: None declared. | **Funding:** None. | **Ethical approval:** Not applicable (hypothesis paper).

© 2026 Martin Hamer, RN (Ret.) | Hamer Glycan Systems | hamerglycans.com | *This hypothesis is offered for scholarly discussion and does not constitute clinical advice.*

Hamer Glycan Systems White Paper Series | © 2026 Martin Hamer, RN (Ret.) | hamerglycans.com